

CLASH: Color Line-Following and Autonomous Stacking Hunt.

Overall flow of the event

PHASE 1 — QUALIFIER ROUND

1. Track & Physical Layout

Parameter	Value
Track color	Black line on white floor
Track width	25 mm
Cube size	50 × 50 × 50 mm (5×5×5 cm)
Cube placement side	Left side of the track

Cube Distance Rule

- The front face of each cube will be placed at *15 cm from the center of the track.*
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2. Color Detection Strips

The robot does **not** detect the cube color directly. It detects the **colored strip** placed on the floor.

Color Strip Dimensions

Parameter	Value
Starting distance from track center	9 cm (90 mm)
Strip length	6 cm (60 mm)
Strip width	3 cm (30 mm)
Placement	Only on right side of the bot

3. Cube Counting System

As the robot moves:

- It must maintain a **running count** of
 - **Red cubes**
 - **Green cubes**
 - **Blue cubes**

Counting is based on detecting the **color strips**, not the cube.

4. Checkpoint Logic (Intersection)

There will be:

- A **full black strip** across the track.
- This acts as the **decision checkpoint**.

When the robot reaches this:

1. Counting stops.
 2. The bot determines **which color has the highest count**.
 3. After the intersection, **three boxes (R, G, B)** will be placed.
 4. **The robot must pick the box corresponding to the maximum-count color.**
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5. Final Placement in Phase 1

After picking the correct box:

1. The robot continues following the line.
2. It detects **broken black lines** (end-of-track pattern).

Broken Line Specification

Component	Dimension
Gap length	10 mm

3. After detecting broken lines, the robot looks for the **FINISH region**:
 - Full-black rectangular region

- 250 mm × 500 mm

Phase 1 Final Action

👉 The robot must carry the *maximum-count cube* to the FINISH zone and drop it there.

PHASE 2 — MAIN EVENT

A complex, undisclosed new map is given. Paths may include:

- Loops
 - T-junctions
 - Intersections
 - Right-angle & acute turns
 - Diode-shaped paths
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Entity Meaning

Color	Meaning	Action
 Red	Bomb Hazard	Must be collected
 Blue	Radioactive Material	Must be collected
 Green	Civilians	Must NOT be collected

STAGE 1 — SCANNING RUN

The robot must:

- Navigate the entire track once.
 - Identify and count all Red, Blue, and Green boxes.
 - Use the same **strip dimensions** and **cube distances** as Phase 1.
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STAGE 2 — COLLECTION & DELIVERY RUN

After the scanning run:

1. When the robot reaches the **broken black line region**, it performs a **180° turn**.
2. It may:
 - return to the starting area, **or**
 - immediately begin collecting boxes near the broken line area.

Collection Rules

- Bot must pick only **Red** and **Blue** cubes.
 - Green cubes must not be disturbed.
 - After picking a cube, the bot will again see the broken lines and must:
 - Cross them
 - Move forward to reach an **intersection**
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FINAL INTERSECTION LOGIC (MANDATORY HARD CODING)

At the intersection after the broken lines:

Turn Direction	Meaning
Left turn	Go to Red drop zone
Right turn	Go to Blue drop zone
Straight	Go to FINISH region

Drop Zone Identification

Each drop zone is marked using **rectangular colored strips** on the **right side of the bot**.

END CONDITION

In Phase 2, after delivering all required boxes the bot should reach the finish zone
